ECE Sizing Calculator

Elastic Cloud Enterprise Deployment Planner

User Guide & Calculation Reference

v1.0.9 · Expedient Technology Solutions

This guide explains how to use the ECE Sizing Calculator to plan Elastic Cloud Enterprise deployments, interpret results, and understand the formulas behind every number.

1. Overview & Quick Start

The ECE Sizing Calculator helps you plan the right hardware and Elasticsearch cluster configuration for an Elastic Cloud Enterprise deployment. It takes your data characteristics as inputs — ingest rate, retention, use case — and outputs RAM, storage, node counts, and shard health guidance sized to Elastic's published best practices.

Live URL: ece-sizer.pplx.app · No login required · Works in any modern browser

Quick Start (5 steps)

- Select a Use Case Preset (SIEM, APM, Security Analytics, Enterprise Search, or Custom).
- Choose an Architecture Preset (Elastic Standard 3-tier, Expedient Optimized, or Minimal).
- Enter your Daily Ingest (GB/day) and Availability Zones.
- Set retention days for each active tier (Hot / Warm / Cold / Frozen).
- Review the Results section — RAM, Storage, Object Store, and Shard Health.

Tip: Start with a preset and adjust only what you know. The defaults are based on Elastic's official sizing guidelines and are safe starting points.

Use Case Presets

Preset	Best For	Key Defaults Set
SIEM	Security information & event management, log ingestion	Hot 7d, Warm 23d, Cold 60d; Hot ratio 1:30
APM	Application performance monitoring, traces, metrics	Hot 3d, Warm 14d; higher replica count
Security Analytics	Threat hunting, detection engineering, Elastic Security	Hot 14d, Warm 30d, Frozen 365d; frozen tier enabled
Enterprise Search	App search, website search, document retrieval	Non-rolling index type; cold/frozen disabled
Custom	Any workload not covered above	All fields left at last-used values for manual tuning

Architecture Presets

Preset	Description	When to Use
Elastic Standard	3 AZs, 1 replica, dedicated masters enabled	Production — full HA, Elastic recommended baseline
Expedient Optimized	3 AZs, 1 replica, masters auto-scaled, coordinator enabled	Expedient-hosted ECE — tuned for multi-tenant allocator environment
Minimal	1 AZ, 0 replicas, no dedicated masters	Dev/test only — no fault tolerance

2. Input Fields Reference

All primary inputs are in the top card. Fields update results in real time as you type.

Field	What to Enter	Notes
Client / Project Name	Free text label	Included in Export JSON filename
Daily Ingest (GB/day)	Uncompressed source data volume per day	Use pre-compression size. Elastic typically achieves 2–4× compression.
Availability Zones	1, 2, or 3	3 AZs required for production HA. Replicas are distributed across zones.
Hot Retention (days)	Days data stays on hot (SSD) nodes	Typical: 3–14 days for logs; 30+ days for search data.
Hot Replicas	Number of replica shards per primary	1 replica = full copy of data (doubles storage). Minimum 1 for HA.
Warm Retention (days)	Additional days on warm (HDD) tier after hot	Enable warm toggle first. 0 = warm tier not used.
Cold / Frozen Retention	Days in cold (searchable snapshot) or frozen (partial mount) tiers	Frozen tier uses object storage — very low cost per GB.
Dedicated Masters	Auto (recommended) or manual count + RAM	Auto picks 4/8/16 GB based on node count and shard count.
Coordinator Nodes	Optional — count and RAM	Recommended for deployments with >10 hot nodes or heavy Kibana usage.

3. Advanced Ratios & Tuning

The Advanced section is organized into five groups. Most users can leave these at defaults — they represent Elastic's published planning guidelines. Adjust only when you have specific hardware specs or measured workload data.

Note on heuristics: Overhead multipliers and the CPU ingest rule are planning heuristics used in Elastic sizing workshops — not officially published constants. Actual values vary by workload. Treat them as conservative starting points.

Group / Field	Default	What It Controls
RAM : Storage Ratios		
Hot (1:X)	1:30	GB of RAM per GB of indexed data on hot nodes. Higher = more storage-dense but slower.
Warm (1:X)	1:160	Warm nodes are force-merged and read-only — much higher ratio is safe.
Cold (1:X)	1:160	Cold uses the same hardware profile as warm (1:160). Storage savings come from searchable snapshots eliminating replica copies — not from a lower RAM ratio.
Frozen (1:X)	1:1500	Frozen nodes hold only metadata and a small local cache. Dataset lives in object store. Elastic published ratio: 1:1500.
Overhead & Watermark		
Hot Overhead Multiplier	1.5×	Lucene segment overhead heuristic for hot tier. Conservative planning default. Actual overhead varies with field mappings and merge state (typically 1.4–1.7×).
Warm Overhead Multiplier	1.25×	Lower than hot because warm data is force-merged (fewer segments). Typical range: 1.15–1.30×.
Cold/Frozen Overhead	1.1×	Snapshot data is fully merged — minimal overhead.
Disk Watermark Headroom	15%	Elasticsearch blocks writes at 85% disk usage by default. This buffer ensures you never hit it.
Shard & Index Sizing		
Target Shard Size (GB)	30 GB	Elastic recommends 10–50 GB. Used to compute primary shard count.
System Indices Shards	250	Baseline for .kibana, .security, and other built-in indices.
Indices/Day	1	Distinct data streams writing per day. Increase for multi-source deployments.
ILM Rollover		

Rollover Condition	Age-based	Age-based: one index per day. Size-based: ILM rolls when index hits max size.
Max Index Size (GB)	50 GB	(Size-based only) Elastic ILM max_primary_shard_size threshold.
Snapshots		
Snapshot Retention (days)	7	Days of nightly snapshot backups kept in object store. Calculator shows conservative upper bound — actual storage is typically 30–60% lower due to incremental deduplication.
Index Type	Rolling	Rolling (logs/metrics): daily delta only. Non-rolling (search): adds full initial snapshot.
Infrastructure		
Hot vCPU / Node	8	vCPUs assigned to each hot node. Used for CPU ingest capacity check (planning heuristic).
ECE Allocator Count	14	Number of allocator hosts. Each uses 12 GB for ECE control plane (per Elastic documentation).
Allocator RAM / Host (GB)	1412	Physical RAM per allocator. ECE control plane overhead (12 GB) is subtracted from usable tenant RAM.
Nodes/Zone (per tier)	1	Nodes per AZ for hot/warm/cold/frozen. Affects total shard capacity ceiling.
Frozen Cache/Node (GB)	1000 GB	Local SSD cache for frozen tier. Elastic guidance: 1–10% of remote dataset; 5% is the recommended starting point.

4. Results & KPIs

The Results section updates instantly as you change inputs. It shows five key output areas:

KPI	What It Shows	How to Interpret
Total RAM	Sum of all Elasticsearch node RAM across all tiers + masters + coordinators	Sub-line shows tenant RAM utilization % of total allocator capacity (12 GB ECE overhead per allocator deducted).
Total Storage	Hot + Warm + Cold on-node storage including replicas, overhead, and watermark buffer	This is the raw disk capacity needed on allocator hosts — not object store.
Object Store	Frozen tier data + snapshot retention in S3/Blob/GCS	Low-cost storage tier. Frozen data is fetched on demand — not stored locally. Snapshot estimate is a conservative upper bound.
Frozen Cache Health	Cache % of frozen data — ok (≥10%), warn (5–10%), bad (<5%)	Below 5% risks cache thrashing. Elastic guidance: 5% minimum starting point. Increase toward 10% for high cache-hit workloads.
CPU Hint	Ingest GB/hr vs. vCPU capacity check using 4 vCPU/GB-hr planning heuristic	Yellow/red means hot nodes may be CPU-bound at peak ingest. Not official Elastic guidance — actual CPU needs vary by workload complexity.

Node Breakdown Table

Below the KPIs, a table shows each component — Hot, Warm, Cold, Frozen, Master, Coordinator — with its RAM per node, node count (per zone × AZs), total RAM, and storage contribution.

> ECE Instance Size Snapping: The calculator automatically snaps each node's RAM to the nearest valid ECE instance size (data.default profile): 1, 2, 4, 8, 16, 32, or 64 GB. This ensures the output reflects what ECE can actually allocate.

5. Shard Model

The Shard Model section provides a health check on your shard configuration against Elastic's best practices. It is the most actionable section for detecting under- or over-sharded deployments before they cause problems in production.

Shard Health Banner

A color-coded banner summarizes overall shard health:

- Green (OK): All shard density and count checks pass.
- Yellow (Warning): One or more thresholds approaching limit — review recommendations.
- Red (Critical): Cluster shard limit or density ceiling exceeded — action required.

Shard Table

The table shows per-tier shard counts broken down into:

Column	Meaning
Primary Shards	Derived from: ceil(daily ingest on-disk ÷ target shard size) × indices/day × retention days
Replica Shards	Primary shards × replica count
Total Shards	Primary + replica
Shards/Node	Total shards ÷ nodes in tier. Hard limit: 1,000/node (Elastic 8.3+ cluster.max_shards_per_node default).
Ceiling	Max shards the tier can host: hot/warm = 1,000/node; cold = 1,000/node; frozen = 3,000/node.
Heap/Node	Node RAM ÷ 2, capped at 31 GB (JVM limit)
Utilization Bar	Shards/node as % of the per-node ceiling. Red >100%, yellow >75%.

Cluster-Wide Checks

- Total cluster shards vs. dynamic limit: Elastic 8.3+ guideline is 1,000 shards per non-frozen data node. The calculator computes this limit dynamically — e.g., a cluster with 12 non-frozen nodes has a limit of 12,000 shards. This replaced the older fixed 50,000 soft limit in Elasticsearch 8.3.
- Hot+Warm shard density: Cluster-wide shards across hot and warm tiers combined divided by total hot+warm JVM heap. Should stay below 20 shards/GB of heap.

Recommendations Panel

When issues are detected, the Recommendations panel shows actionable guidance. Possible recommendations include:

- Increase target shard size (shards too small / too many)
- Reduce indices/day or use data streams (too many index groups)
- Add hot/warm nodes per zone (density too high)

- Reduce retention (too many shards accumulating)
- Increase replica count (under-replicated for HA)
- Frozen cache too small (less than 5% of frozen data)

6. How the Calculations Work

This section documents every formula the calculator uses. All variables correspond directly to input field IDs in the calculator.

6.1 ECE Instance Size Snapping

Before computing heap or shard ceilings, node RAM is snapped to the nearest valid ECE instance size (data.default instance configuration):

```
Valid sizes (GB): 1, 2, 4, 8, 16, 32, 64
```

The calculator picks the smallest valid size >= the computed node RAM. For nodes requiring more than 64 GB, it rounds up to the next multiple of 64 GB. This ensures all outputs are achievable on a real ECE deployment.

6.2 Storage per Tier (Age-Based ILM)

```
hot_data = ingest × hot_days × (1 + hot_replicas) × hot_overhead
warm_data = ingest × warm_days × (1 + warm_replicas) × warm_overhead
cold_data = ingest × cold_days × (1 + cold_replicas) × cold_overhead
hot_storage = hot_data × watermark_multiplier
warm_storage = warm_data × watermark_multiplier
watermark_multiplier = 1 / (1 - watermark_pct/100) [default: 1/0.85 ≈ 1.176]
```

Overhead multipliers are planning heuristics: hot 1.5×, warm 1.25×, cold/frozen 1.1×. Warm is lower because ILM force-merges indices before transitioning them. Actual overhead depends on field mappings, document size, and merge policy.

6.3 Storage per Tier (Size-Based ILM Rollover)

When Size-based rollover is selected, the hot tier calculation changes:

```
daily_on_disk = ingest × hot_overhead
active_per_stream = ceil(daily_on_disk / max_index_size_gb)
hot_active_indices = active_per_stream × indices_per_day
hot_data = hot_active_indices × max_index_size_gb × (1 + hot_replicas) × hot_overhead
```

This models the fact that with size-based rollover, multiple partially-filled indices may exist simultaneously on hot nodes rather than a single daily index.

6.4 Object Store (Snapshots + Frozen)

```
snapshot_daily = ingest × hot_overhead [rolling index type]
snapshot_storage = snapshot_daily × snapshot_retention_days
# Non-rolling index type adds initial full snapshot:
initial_snapshot = (hot_data + warm_data) [non-rolling only]
frozen_storage = ingest × frozen_days × (1 + frozen_replicas) × cold_overhead
object_store_total = snapshot_storage + initial_snapshot + frozen_storage
```

Snapshot storage note: The formula computes a conservative upper bound. Elasticsearch snapshots are incremental and deduplicate unchanged Lucene segments. Actual object storage is typically 30–60% lower than this estimate.

6.5 RAM per Tier

```
hot_ram = hot_data / hot_ratio
warm_ram = warm_data / warm_ratio
cold_ram = cold_data / cold_ratio [cold_ratio default: 160]
frozen_ram = frozen_obj / frozen_ratio [frozen_ratio default: 1500]
node_ram = tier_ram / az_count / nodes_per_zone
snapped_node_ram = snapToECE(node_ram) [snap to valid ECE instance size]
heap_per_node = min(snapped_node_ram / 2, 31) [JVM 31 GB heap cap]
```

Cold tier RAM note: Cold uses the same hardware profile as warm (1:160 default). The apparent cost savings of cold vs. warm come from eliminating replica copies via searchable snapshots — not from a lower RAM ratio.

6.6 Dedicated Master RAM (Auto)

```
if total_data_nodes > 30 OR total_shards > 20,000: master_ram = 16 GB
elif total_data_nodes > 10 OR total_shards > 5,000: master_ram = 8 GB
else: master_ram = 4 GB
```

6.7 Shard Count

```
# Age-based:
pri_shards = ceil(ingest × hot_overhead / target_shard_size) × indices_per_day
# Size-based:
pri_shards = ceil(max_index_size / target_shard_size) × hot_active_indices
hot_shards = pri_shards × hot_days × (1 + hot_replicas) + system_shards
warm_shards = pri_shards × warm_days × (1 + warm_replicas)
cluster_shards = hot_shards + warm_shards + cold_shards + frozen_shards
```

6.8 Shard Density Check (per node)

Elasticsearch 8.3+ enforces a hard per-node shard limit via cluster.max_shards_per_node. The calculator uses this as the ceiling for all tiers:

```
hot_ceiling = 1,000 shards/node [cluster.max_shards_per_node default]
warm_ceiling = 1,000 shards/node
cold_ceiling = 1,000 shards/node [searchable snapshots, no indexing]
frozen_ceiling = 3,000 shards/node [metadata + cache only]
shards_per_node = tier_total_shards / tier_total_nodes
density_pct = shards_per_node / tier_ceiling × 100
```

Note: The older rule of '20 shards per GB of JVM heap' was deprecated in Elasticsearch 8.3 in favor of the flat 1,000/node limit. The calculator still shows heap-based density as a secondary metric for hot+warm tiers combined.

6.9 Cluster Shard Limit

```
non_frozen_nodes = hot_nodes + warm_nodes + cold_nodes
cluster_shard_limit = non_frozen_nodes × 1,000
warn_threshold = cluster_shard_limit × 0.80
```

The limit scales with cluster size: a 12-node non-frozen cluster has a 12,000 shard limit. This replaced the older fixed 50,000 soft limit in Elasticsearch 8.3.

6.10 ECE Allocator Overhead

```
ECE_CTRL_PLANE_GB = 12 [per allocator host — Elastic documentation]
allocator_overhead_gb = allocator_count × 12
allocator_total_gb = allocator_count × allocator_ram_per_host
allocator_tenant_gb = allocator_total_gb - allocator_overhead_gb
tenant_utilization_pct = total_ram / allocator_tenant_gb × 100
```

What this means: The sub-line under Total RAM shows how much of your allocator fleet's usable tenant RAM this deployment consumes. Above 80% is a planning warning — leave headroom for cluster resizing and other tenants.

6.11 CPU Ingest Check (Planning Heuristic)

```
ingest_gb_hr = ingest / 24
vcpus_needed = ingest_gb_hr × 4 [heuristic: ~4 vCPU per GB/hr]
vcpus_available = hot_vcpu_per_node × hot_total_nodes
cpu_ratio = vcpus_needed / vcpus_available
```

Green: ratio < 1.0 · Yellow: 0.5–1.0 (approaching) · Red: >1.0 (CPU-bound risk)

Heuristic caveat: The 4 vCPU/GB-hr rule is a workload-specific planning heuristic, not an officially published Elastic constant. CPU needs vary significantly based on indexing pipeline complexity, field mapping count, and ingest processor usage. Use this as an early warning indicator only.

7. Saving, Sharing & Importing

Export JSON

The Export JSON button downloads a complete snapshot of all input field values. The file is named with the client name and timestamp. This is the primary way to save a sizing calculation for later or to hand off to a colleague.

Import JSON

The Import JSON button (blue, next to Export) loads a previously exported file back into the calculator. All fields are restored exactly as they were when the file was exported. The results section scrolls into view automatically after import.

> Version compatibility: The importer supports both the current nested format and older flat export formats from previous versions. Legacy exports will import correctly.

Share URL

The Share URL button copies a link to your clipboard that encodes the current configuration as URL parameters. Anyone with the link can open the calculator with your exact inputs pre-loaded — no file needed. Useful for quick handoffs or embedding in a proposal.

> Note: Share URLs encode inputs only — they do not include client name or notes. Use Export JSON for a full record.

Reset Defaults

The Reset button restores all fields to calculator defaults. This does not affect any previously exported JSON files.

8. Tips & Common Mistakes

Getting Accurate Ingest Numbers

- Use pre-compression source data size, not what you see on disk in Elasticsearch. Elastic compresses data 2–4×, but the calculator accounts for this via the overhead multipliers.
- If you only have disk usage in Kibana, multiply it by 2–3 to estimate source ingest volume.
- For SIEM deployments, EPS (events per second) × average event size (bytes) × 86,400 = GB/day.

Choosing Retention Correctly

- Hot retention should reflect how long data needs fast (sub-second) query response — usually 3–14 days for logs.
- Warm retention covers data that is queried occasionally and can tolerate slightly slower response (seconds). Force-merged and read-only.
- Cold and frozen tiers are for compliance/archival — query performance is not guaranteed.

Shard Sizing

- The 10–50 GB target shard size is an Elastic best practice. Shards below 10 GB create unnecessary overhead. Shards above 50 GB slow recovery times.
- The shard ceiling is 1,000 shards per node (Elasticsearch 8.3+ default). If the shard model shows red, the most common fix is to increase target shard size (reduce shard count) or add nodes per zone (distribute shards across more nodes).
- System indices shards (default 250) account for .kibana, .security, .fleet, etc. In large Elastic Security deployments this may need to be higher (400–600).

Frozen Tier

- Frozen is economical but not free — ensure your object store (S3/Azure Blob/GCS) is provisioned with sufficient capacity.
- Elastic guidance: target 5–10% of frozen data volume as local cache. Below 5% causes excessive object store fetches and slow query times. 5% is the recommended starting point for new deployments.

Cold vs. Frozen: Which to Use

- Cold: Searchable snapshots with full index mounted locally. Lower latency, higher RAM. Good for data queried a few times per week.
- Frozen: Partially-mounted snapshots — most data stays in object store. Very low RAM, higher query latency. Best for compliance archival rarely queried.

Common Mistakes

- Setting retention too high on hot tier: Hot nodes use expensive NVMe SSD — move data to warm after 7–14 days to reduce cost.
- Ignoring the watermark buffer: Without 15% headroom, Elasticsearch will stop indexing when disks fill up. Never plan to use 100% of disk capacity.
- Zero replicas: A 0-replica cluster has no data redundancy — a single node failure causes data loss and index red state.

- Using Minimal preset for production: The Minimal preset (1 AZ, 0 replicas) is for dev/test only. Always use at least 2 AZs with 1 replica in production.
- Underestimating cold RAM: Cold uses the same 1:160 hardware ratio as warm. The storage savings come from searchable snapshots eliminating replicas — plan cold RAM accordingly.

For questions about this calculator or Expedient's Elastic Cloud Enterprise offerings, contact your Expedient Technical Resolution Manager.

Calculator: ece-sizer.pplx.app

9. References & Further Reading

The formulas and thresholds in this calculator are derived from the following Elastic official documentation and published guidance. These sources are the authoritative reference for any value used in the calculator.

Sizing & Hardware Guidance

Source	Description	URL
Elastic hardware recommendations	Elastic's official hardware sizing reference for hot, warm, cold, and frozen tiers. Source for RAM:storage ratios (hot 1:30, warm 1:160, frozen 1:1500) and tier hardware profiles.	https://www.elastic.co/guide/en/cloud/current/ec-hardware.html
Size your deployment	Elastic Cloud sizing guide covering RAM:storage ratios, node sizing, and tier recommendations.	https://www.elastic.co/guide/en/cloud/current/ec-customize-deployment.html
ECE allocator planning	ECE documentation specifying 12 GB of ECE control plane overhead per allocator host.	https://www.elastic.co/guide/en/cloud-enterprise/current/ece-allocators.html
ECE instance configurations	Reference for official ECE data.default instance sizes: 1, 2, 4, 8, 16, 32, 64 GB.	https://www.elastic.co/guide/en/cloud-enterprise/current/ece-instance-configurations.html

Shard Management

Source	Description	URL
Size your shards	Elastic's primary reference for shard sizing guidance: 10–50 GB target shard size, ILM rollover best practices, and the 1,000 shards/node cluster limit.	https://www.elastic.co/guide/en/elasticsearch/reference/current/size-your-shards.html
cluster.max_shards_per_node setting	Elasticsearch 8.3+ setting that enforces 1,000 shards per node as the default hard limit. Replaces the older 50,000 cluster-wide soft limit.	https://www.elastic.co/guide/en/elasticsearch/reference/current/misc-cluster-settings.html#cluster-max-shards-per-node
ILM: Manage the index lifecycle	Index Lifecycle Management documentation covering rollover conditions, max_primary_shard_size, and tier transition policies.	https://www.elastic.co/guide/en/elasticsearch/reference/current/index-lifecycle-management.html

Disk & Memory

Source	Description	URL
Disk-based shard allocation	Documents Elasticsearch's disk watermark defaults: low 85%, high 90%, flood 95%. Source for the 15% watermark headroom buffer used in storage calculations.	https://www.elastic.co/guide/en/elasticsearch/reference/current/disk-usage-watermark.html

Heap sizing and JVM	Elastic recommendation to cap JVM heap at 31 GB to stay below the compressed ordinary object pointer (compressed-oops) threshold used in shard ceiling calculations.	https://www.elastic.co/guide/en/elasticsearch/reference/current/advanced-configuration.html#set-jvm-heap-size
Dedicated master nodes	Elastic guidance on when to use dedicated master nodes and how to size them (4/8/16 GB). Source for the auto-master RAM scaling thresholds.	https://www.elastic.co/guide/en/elasticsearch/reference/current/master-election.html

Searchable Snapshots & Frozen Tier

Source	Description	URL
Searchable snapshots overview	Explains cold and frozen tier architecture — how fully-mounted vs. partially-mounted snapshots work, and why cold tier RAM requirements align with warm tier (1:160).	https://www.elastic.co/guide/en/elasticsearch/reference/current/searchable-snapshots.html
Frozen tier	Frozen tier architecture detail: partial mount strategy, local cache sizing guidance (1–10% of dataset; 5% starting point), and object store interaction.	https://www.elastic.co/guide/en/elasticsearch/reference/current/data-tiers.html#frozen-tier
Snapshot and restore	Documents incremental snapshot behavior and segment deduplication — basis for the note that actual snapshot storage is typically 30–60% below the calculator's upper bound.	https://www.elastic.co/guide/en/elasticsearch/reference/current/snapshot-restore.html

About this document: This guide is generated from the same codebase as the calculator at ece-sizer.pplx.app. Formula corrections and new features are reflected in updated PDF versions. Current version: v1.0.9.